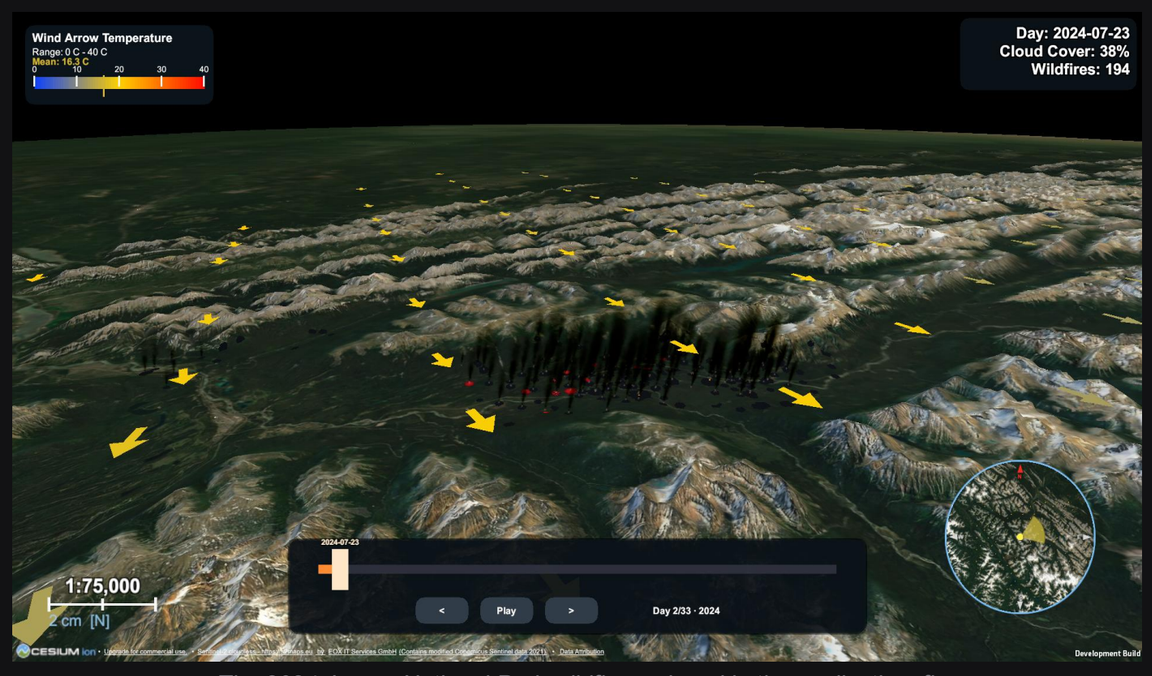
DATA-DRIVEN WILDFIRE SPREAD MODELING USING EARTH OBSERVATION DATA WITH INTERACTIVE 3D VISUALIZATION IN UNITY

DIPLOMA THESIS Marcelo Alves Rigoto · Copernicus Master in Digital Earth



Main menu of the Unity application.



The 2024 Jasper National Park wildfire replayed in the application: fire and smoke on streamed 3D terrain, wind arrows coloured by temperature, one day per step.

OBJECTIVES

- Visualize wildfire extent and temporal progression from MODIS satellite fire detections.
- Implement the full pipeline without paid licences or proprietary GIS platforms, reproducible by non-programmers.
- Let users identify spatial and temporal patterns in an immersive, day-by-day 3D time lapse.
- Establish game engines, and Unity3D specifically, as viable geovisualization platforms beyond entertainment.

METHODOLOGY

01 ACQUIRE
FIRMS detections and Open-Meteo weather, fetched and exported as shapefile (Unity Desktop App) and JSON (WebGL).

→ **02 GEOREFERENCE**
Every detection anchored to a Cesium globe and snapped onto streamed 3D terrain by raycast.

→ **03 ANIMATE**
Detections clustered, scaled by fire radiative power (Unity Desktop App), replayed day by day with wind and cloud.

→ **04 DEPLOY**
One pipeline, two builds: a desktop reference application and a browser version, no installation.

→ **05 EVALUATE**
Counterbalanced study, 68 participants: comprehension, usability and ratings against two tools.

UNITY APPLICATION



TERRAIN
Streamed elevation and satellite imagery

BURN SCARS
Draped on the surface, accumulating each day

FIRE MARKERS
Scaled by fire radiative power

SMOKE
Columns leaning with the recorded wind

WIND
Arrow grid coloured by temperature

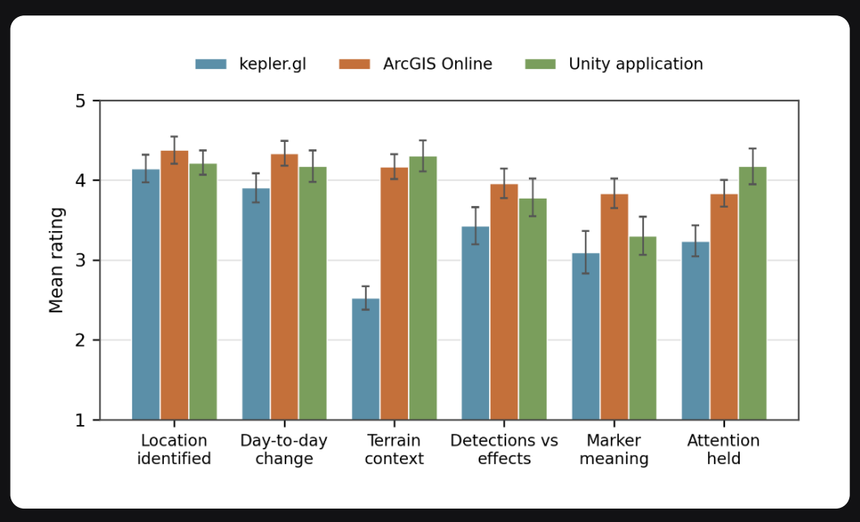
INTERFACE
Timeline, counters, legend and minimap

RESULTS · 68 PARTICIPANTS · UNITY APPLICATION VS KEPLER.GL AND ARCGIS ONLINE

Terrain is a precondition, not decoration.

96-100%
Of the participants read the landscape correctly using a tool that renders 3D terrain

9.5%
Did so without it, and almost every wrong answer was confident that the terrain had no elevation



Unity score the highest with terrain and attention.

DISCUSSION

Communication and usability separate cleanly: the application matches the commercial platform on what it conveys while trailing on ease of operation, an onboarding problem localised in camera navigation.

The scene under-reports the record through a stricter runtime confidence filter. Displaying recorded next to shown counts, and a fire-power legend, are the immediate next steps.

Explore it in your browser
beamish-gelato-46180a.netlify.app

Source code and pipeline
github.com/marcelorigoto1/wildfire-visualization

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